

GURAYAT, Saudi Arabia

WMO: 403600

Lat: 31.412N

Lon: 37.279E

Elev: 1672

StdP: 13.83

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.7	34.1	8.9	9.2	60.3	13.8	11.8	60.5	28.7	52.4	23.8	52.7	3.2	270	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.4	106.0	65.0	103.8	64.9	100.8	64.2	71.7	92.1	69.5	89.8	67.9	88.3	9.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	101.3	81.4	62.9	91.4	77.4	61.1	85.6	75.9	36.6	91.1	34.6	89.7	33.2	87.8	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
25.6	22.8	20.6	DB	25.7	110.9	4.3	2.4	22.6	112.6	20.1	114.0	17.7	115.4	14.5	117.1	
			WB	23.2	73.3	4.3	4.1	20.1	76.3	17.6	78.7	15.2	81.0	12.1	84.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	68.6	48.1	52.1	59.5	68.0	76.7	82.1	85.7	80.8	72.6	59.8	50.6
		DBStd	14.49	4.50	5.10	6.43	7.16	6.28	4.58	3.83	3.94	4.53	5.81	4.77
		HDD50	178	88	31	3	0	0	0	0	0	0	6	50
		HDD65	1763	525	362	197	49	1	0	0	0	9	173	447
		CDD50	6952	28	90	297	541	828	963	1108	923	699	301	68
		CDD65	3060	0	0	26	139	364	513	643	641	473	243	0
		CDH74	41036	0	25	434	1807	4851	7117	9151	8868	5958	2597	225
		CDH80	24171	0	2	139	775	2662	4387	5916	5675	3453	1126	36
(14)	Wind	WSAvg	9.3	6.8	7.9	9.5	9.9	10.6	12.2	13.3	12.3	10.2	7.6	5.7
(15)	Precipitation	PrecAvg	2.0	0.5	0.2	0.3	0.2	0.1	0.0	0.0	0.0	0.2	0.1	0.4
(16)		PrecMax	5.3	2.9	1.0	1.4	1.4	1.1	0.0	0.0	0.0	0.1	0.6	2.2
(17)		PrecMin	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
(18)		PrecStd	1.1	0.6	0.2	0.4	0.4	0.2	0.0	0.0	0.0	0.4	0.2	0.5
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	69.4	77.3	90.7	98.1	104.2	107.6	109.7	109.5	106.0	97.0	84.2	73.1
(20)		0.4% MCWB	51.3	52.8	58.0	61.2	62.5	64.3	66.5	68.1	65.1	63.3	57.7	53.6
(21)		2% DB	64.6	71.7	84.0	92.3	100.1	104.0	105.9	105.7	101.4	93.4	80.2	67.9
(22)		2% MCWB	49.7	51.4	55.9	58.6	61.7	63.3	64.6	66.3	64.3	62.3	57.2	51.8
(23)		5% DB	61.5	67.8	78.6	88.1	96.6	100.6	102.8	102.6	98.2	89.8	76.2	64.4
(24)		5% MCWB	48.3	50.4	53.9	57.5	60.7	63.4	64.9	65.6	63.8	61.3	56.0	50.5
(25)		10% DB	59.0	64.4	73.8	84.3	93.1	97.7	100.4	100.2	95.1	86.3	72.7	61.3
(26)		10% MCWB	47.4	49.2	52.2	56.6	59.9	62.9	64.3	65.1	63.3	60.8	55.1	49.0
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	53.6	55.8	60.2	63.1	67.6	74.4	74.6	75.8	72.3	67.9	62.0	56.7
(28)		0.4% MCDB	63.1	68.5	80.8	85.3	92.7	96.0	98.2	96.2	89.5	85.1	74.8	67.3
(29)		2% WB	51.2	53.5	57.5	60.9	64.3	68.4	70.9	71.5	69.1	65.2	59.8	53.5
(30)		2% MCDB	61.3	67.9	77.7	85.7	90.5	91.2	93.3	91.0	87.1	83.8	72.6	64.0
(31)		5% WB	49.5	51.7	55.1	59.1	62.5	65.9	68.3	69.6	67.1	63.6	58.3	51.6
(32)		5% MCDB	59.1	64.3	74.7	83.6	89.4	90.8	89.9	89.2	86.1	82.3	70.9	62.3
(33)		10% WB	48.2	50.1	53.4	57.3	61.1	64.3	66.9	68.1	65.7	62.3	56.6	49.9
(34)		10% MCDB	57.7	62.2	72.2	81.7	88.2	89.8	89.9	87.9	85.6	80.7	69.3	59.5
(35)	Mean Daily Temperature Range	MDBR	22.2	23.7	25.8	27.7	29.4	30.8	30.4	30.0	29.0	26.8	24.4	22.9
(36)		5% DB MCDBR	26.0	28.9	32.2	31.8	32.2	32.9	31.8	31.2	31.3	29.7	27.7	26.0
(37)		5% WB MCWBR	14.4	13.9	13.3	11.9	10.7	8.7	7.8	7.5	8.5	10.3	12.7	14.5
(38)		5% DB MCDBR	23.2	25.2	28.5	29.6	29.4	31.0	30.2	30.4	28.5	25.8	22.7	22.9
(39)		5% WB MCWBR	13.8	13.4	12.8	12.0	10.6	10.0	8.8	8.5	8.4	9.4	10.8	13.4
(40)	Clear-Sky Solar Irradiance	taub	0.348	0.376	0.424	0.485	0.483	0.408	0.410	0.415	0.408	0.436	0.383	0.354
(41)		taud	2.237	2.146	1.994	1.843	1.875	2.114	2.128	2.120	2.124	2.028	2.146	2.221
(42)		Ebn at Noon	273	276	271	259	259	279	277	275	272	252	257	264
(43)		Edh at Noon	36	43	54	66	64	50	49	49	47	48	39	35
(44)	All-Sky Solar Radiation	RadAvg	1084	1365	1787	2119	2400	2648	2556	2342	2026	1577	1205	1013
(45)		RadStd	48	82	69	69	98	30	47	43	59	61	56	43

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+259	N/A
(47)	Regional (11 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+257	+171

Nomenclature: See separate page